MIL-D-21524A (Wep)
11 August 1961

Superseding
MIL-D-21524 (Weps)
17 November 1960

# MILITARY SPECIFICATION
# FOR
# DETONATOR ASSEMBLY

**This Specification has been approved by the Bureau of Naval Weapons, Department of the Navy**

## 1. SCOPE

1.1 Scope - This specification states the requirements for a detonator assembly used to fire a shaped metal-clad, igniter fuse.

## 2. APPLICABLE DOCUMENTS

2.1 Documents - The following documents of the issue in effect on the date of invitations for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner |
| PPP-B-636 | Boxes, Fiberboard |
| PPP-B-676 | Boxes, Setup, Paperboard |

MILITARY

| | |
|---|---|
| MIL-A-148 | Aluminum Foil |
| MIL-S-901 | Shockproof Equipment, Class HI (High-Impact), Shipboard Application, Tests for |
| MIL-E-5272 | Environmental Testing, Aeronautical and Associated Equipment, General Specification for |

FSC 1375

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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-167 | Mechanical Vibrations of Shipboard Equipment |
| MIL-STD-350 | Jolt Test for Use in Production of Fuzes |
| MIL-STD-352 | Forty (40) Foot Drop Test for Use in Production of Fuzes |
| MIL-STD-354 | Temperature and Humidity Test for Use in Production of Fuzes |
| MIL-STD-356 | Salt Spray (Fog) Test for Use in Production of Fuzes |

COMMERCIAL

American Table of Distances

DRAWINGS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS, Code Ident 10001

| | |
|---|---|
| BuWep LD 267280 | Test Set, Mark 238 Mod 0 (Bridge Resistance) |
| BuWep LD 267390 | Test Set, Mark 160 Mod 1 |
| BuWep LD 512330 | Detonator Assembly |
| BuWep Dwg 2009330 | Clip |
| BuWep Dwg 2016924 | Electric Detonator Assembly |
| BuWep Dwg 2016961 | Plug with Bridge Wire |
| BuWep SA-2586902 | Detonator Test Equipment - Details and Assemblies |



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PUBLICATIONS

NAVY DEPARTMENT

| | |
|---|---|
| NAVWEPS OP 400 | General Instructions for Design, Manufacture, and Inspection of Naval Ordnance Equipment |
| NAVWEPS OP 2230 | Workmanship and Design Practices for Electronic Equipment |
| NAVORD INST 4355.1 | Equipment for Use by Government Inspectors in Final Inspection of Naval Ordnance Items |
| NAVORD INST 4355.28 | Bureau of Ordnance Test Calibration Program |
| NAVORD OD 6852 | Test Set Mark 160 Mod 1 |
| NAVORD OD 6854 | Test Set Mark 238 Mod 0 |

ARMED SERVICES

Armed Services Explosives Safety Board

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications - The following documents, to the extent stated herein, form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

| | |
|---|---|
| 49 CFR 71-78 | Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles |

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)



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CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

## 3. REQUIREMENTS

3.1 Description - The detonator assembly consists of an electrically initiated actuating charge, a detonating charge, and a bridge plug assembled in an encapsulating cup and is used to fire metal-clad igniter fuse.

3.2 General Requirements

3.2.1 Preproduction Samples - Unless otherwise specified in the contract or order, preproduction samples of the detonator assembly shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification (refer to paragraph 4.2.1). The requirement for a preproduction sample may be waived by the procuring activity if the contract is awarded to an agency which is currently producing the detonator described by this specification or which has successfully completed a production order within the preceding 6 months. In either event, such a waiver is not automatic, but must be specifically requested.

3.2.2 Materials - Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the detonator assembly shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BuWep LD 512330.

3.2.3 Finishes, Coatings, Construction, and Dimensions - Finishes, coatings, construction, and dimensions for the detonator assembly shall be in accordance with the applicable drawings, specifications, and publications listed in BuWep LD 512330.



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3.2.4 Marking - Marking of assemblies, subassemblies, and parts shall be in accordance with applicable drawings listed in BuWep LD 512330.

3.2.5 Interchangeability - Individual assemblies and piece parts of assemblies fabricated in agreement with their applicable drawings and specifications shall be interchangeable with like numbered parts.

3.2.6 Gages - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirement shown on the applicable drawings. If so stipulated in the contract, requisition, or order, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.7 Request for Deviations - The procedures for requesting deviations from this specification or any specification, publication, material, or process specified herein, shall be as set forth in NAVWEPS OP 400.

3.2.8 Conflicting Requirements - In event of conflict between this specification and any specification, publication or drawing referenced herein, this specification shall govern.

3.2.9 Tolerances - Unless otherwise specifically noted, tolerances and accuracies stated herein are absolute.

3.3 Performance Requirements

3.3.1 Bridge Plug

3.3.1.1 Insulation Resistance - The insulation resistance of the bridge plug shall be a minimum of 50 megohms at 500-volts DC. Inspection and testing procedure shall be in accordance with paragraph 4.5.4.2.1.



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3.3.1.2 Bridge Circuit Resistance - The bridge plug shall have a direct-current resistance as specified on BuWep Dwg 2016961. Inspection and testing procedure shall be in accordance with 4.5.4.2.2.

3.3.2 Assembly - When assembling the primer holder on the bridge wire assembly, care shall be exercised to avoid injury to the bridge wire during the assembly operation. The force required to seat the bridge wire assembly in the primer holder shall be not less than 10 pounds, and the primer holder shall be tight on the plug. Acceptance shall be in accordance with 4.5.4.3.

3.3.3 Detonator Assembly

3.3.3.1 Insulation Resistance - The detonator assembly shall have an insulation resistance of not less than 50 megohms, measured between the shorted prongs and the outside of the cup, under a potential of 500-volts DC. Testing and acceptance shall be in accordance with 4.5.4.4.1.

3.3.3.2 Bridge Circuit Resistance - The electrical circuit of the detonator (before shunting leads) shall have a direct-current resistance as specified on BuWep Dwg 2016961. Testing and acceptance shall be in accordance with 4.5.4.4.2.

3.3.3.3 Air and Watertightness - The detonators shall show no evidence of leakage when subjected to the air and watertightness test of 4.5.4.4.3.

3.3.3.4 Function/Output - The detonator shall be subjected to the test of 4.5.4.4.4 and shall function with an output as follows:

(a) The average crater depth of each sample lot shall exceed 0.050 inches.

(b) The minimum crater depth shall be 0.040 inches.

3.3.3.5 Detonator Time - The detonator shall be subjected to the test of 4.6.7 and shall function within 600 microseconds after voltage is applied.



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3.3.3.6 Sensitivity

3.3.3.6.1 Current - The detonator shall be subjected to the test of 4.6.8.1, shall not fire, and thereafter shall meet all other performance requirements of this specification.

3.3.3.6.2 Radio Frequency - The detonator shall be subjected to the test of 4.6.8.2, shall not fire, and thereafter shall meet all other performance requirements of this specification.

3.4 Environmental Conditions

3.4.1 Vibration

3.4.1.1 Vibration, Operating Condition - The detonator shall be subjected to the shipboard vibration test specified in 4.6.1.1 and thereafter shall be capable of meeting the requirements of 3.2.3, 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.3.4, and 3.3.3.5.

3.4.1.2 Vibration, Shipping Condition - The detonators, packaged and packed in accordance with section 5 herein, shall withstand the following vibration conditions, and thereafter shall be capable of meeting all performance requirements of 3.2.3, 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.3.4, and 3.3.3.5.

VIBRATION CRITERIA (MAX FOR PACKING)

| Type of Transportation | Vertical cps | Vertical g | Longitudinal cps | Longitudinal g | Lateral cps | Lateral g |
|---|---|---|---|---|---|---|
| Truck | 200 | 2.5 | 150 | 1.5 | 200 | 1.0 |
| | 10 | 1.0 | 10 | 1.0 | 10 | 1.0 |
| Rail | 60 | 2.0 | 60 | 1.0 | 30 | 1.0 |
| | 10 | 1.0 | 10 | 0.5 | 10 | 0.5 |
| Aircraft | 60-100 | 1.0 | 60-100 | 1.0 | 60 | 1.0 |
| | 100-200 | 0.5 | | | 100 | 0.5 |
| Ship (surface) | 10 | 0.5 | Negligible | | 10 | 0.25 |
| | 30 | 1.5 | | | 30 | 0.75 |



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3.4.2 Shock

3.4.2.1 Shock, Operating Condition - The detonator shall be subjected to the shock test of 4.6.2.1 and thereafter shall be capable of meeting all requirements of 3.2.3, 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.3.4, and 3.3.3.5.

3.4.2.2 Shock, Shipping Condition

3.4.2.2.1 Impact - The detonators, packaged and packed in accordance with 5.1 and 5.2, shall be tested by the procedure specified in 4.6.2.2.1, shall show no damage, and thereafter shall meet the requirements of 3.2.3, 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.3.4, and 3.3.3.5.

3.4.2.2.2 Corner Drop - The detonators, packaged and packed in accordance with 5.1 and 5.2, shall withstand the test specified in 4.6.2.2.2, shall show no damage, and thereafter shall meet the requirements of 3.2.3, 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.3.4, and 3.3.3.5.

3.4.3 Temperature and Humidity - The detonator shall be subjected to the temperature and humidity test of 4.6.3 and thereafter shall be capable of meeting the performance requirements of 3.2.3, 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.3.4, and 3.3.3.5.

3.4.4 Corrosive Atmosphere - The detonator shall withstand salt spray (fog) atmosphere as specified in 4.6.4 for 48 hours and for 96 hours. After the 48-hour test, the detonator shall show no damage and shall meet all performance requirements of 3.2.3, 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.3.4, and 3.3.3.5. After the 96-hour test the detonator shall be safe for handling and disposition.

3.4.5 Fungus - The detonator shall be subjected to the fungus test of 4.6.5, shall not show any evidence of the support of fungal growth, and shall subsequently be capable of meeting the requirements of 3.2.3, 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.3.4, and 3.3.3.5.



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3.4.6 Safety Requirements

3.4.6.1 Durability - The detonator shall withstand the jolt test of 4.6.6.1 without firing and shall subsequently be safe for handling and disposition.

3.4.6.2 Forty-Foot Drop - The detonator shall withstand the forty-foot drop test of 4.6.6.2 without firing and shall subsequently be safe for handling and disposition.

3.4.6.3 Four-Foot Drop - The detonator shall be subjected to the four-foot drop test of 4.6.6.3, shall not detonate, and subsequently shall be capable of meeting the requirements of 3.2.3, 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.3.4, and 3.3.3.5.

3.5 Cleanup - Prior to and after final assembly all parts shall be thoroughly cleaned of all foreign matter.

3.6 Workmanship - The detonator assembly shall be constructed and finished in a manner to assure compliance with this specification. NAVWEPS OP 2230 shall be used as a guide for determining acceptable workmanship of electrical and electronic parts.



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## 4. QUALITY ASSURANCE PROVISIONS

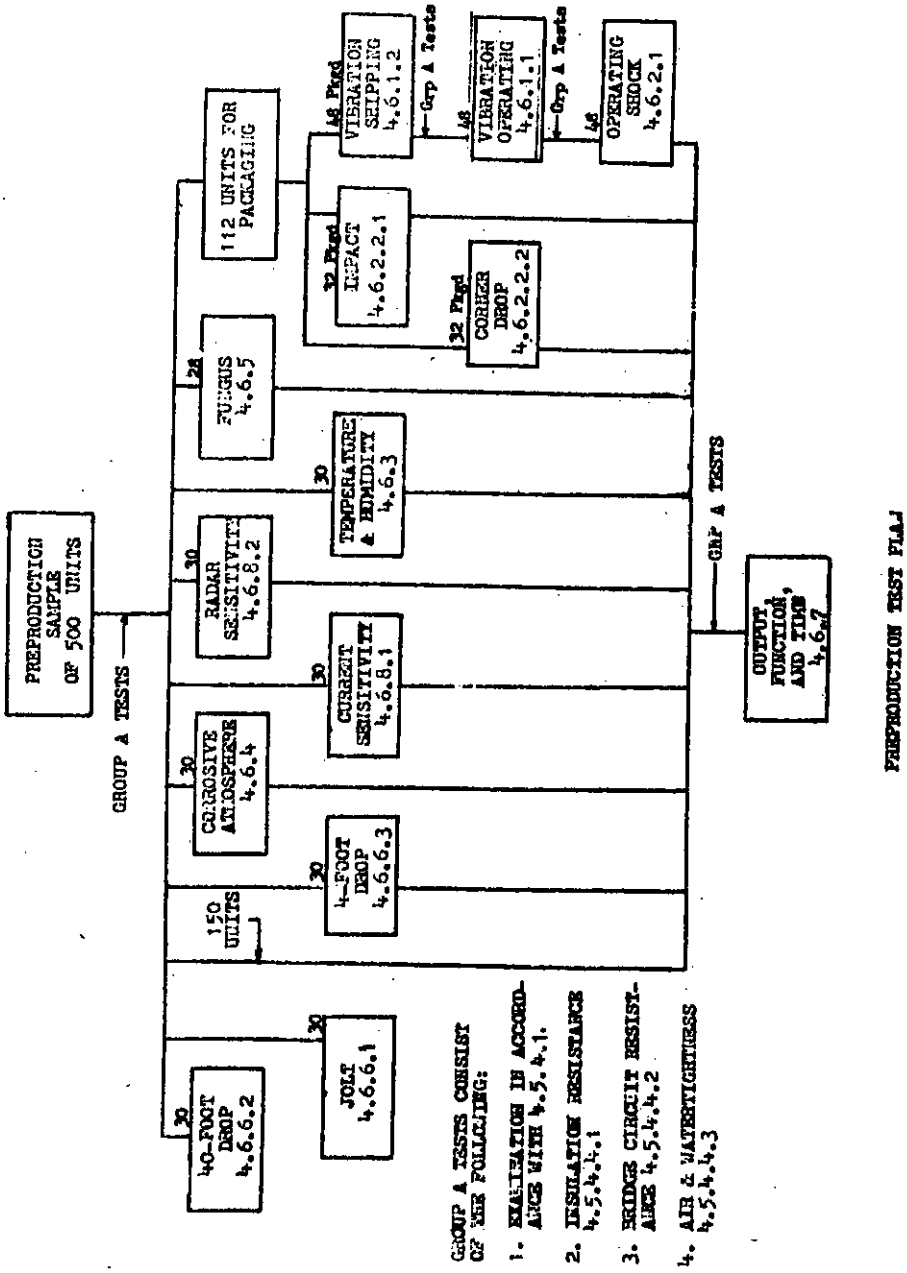
4.1 Contractor's Inspection - Unless otherwise specified herein, the contractor is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the contractor may utilize his own facilities or any laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Government Inspection - The Government Inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract, pertinent drawings, and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in NAVWEPS OP 400. Unless otherwise specified, production tests of final assemblies shall be on a 100-percent basis. No sampling, except as specified, is permitted for acceptance inspection purposes. Only those assemblies which pass final inspection shall be accepted by the Navy; a rejected assembly may be reworked and submitted in accordance with NAVWEPS OP 400.

4.2.1 Preproduction Sample - The preproduction sample specified in 3.2.1 shall be submitted to the Government Inspector for preproduction tests at an activity designated in the contract, or by the Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor, prior to approval by the contracting activity or the completion of tests on the preproduction samples, shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order. The preproduction sample shall be allocated in accordance with Figure 1.



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PREPRODUCTION TEST PLAN

Figure 1



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4.2.2 Preproduction Sample Size - Unless otherwise specified, the preproduction sample shall consist of 500 detonator assemblies.

4.2.3 Lot - The term "lot" shall mean "inspection lot", a collection of like units of products submitted by a supplier for Government inspection and, unless otherwise specified, shall consist of 650 or 700 detonator assemblies which have successfully passed production tests performed by the supplier. Of three successive lots, the first two shall consist of 650 detonator assemblies; the third lot shall consist of 700 detonator assemblies.

4.2.4 Production and Periodic Production Samples - From each succeeding lot, a random sample of 150 detonator assemblies from 650 or 200 detonator assemblies from 700 will be selected by the Government Inspector for additional testing to determine compliance of the samples with the requirements of the contract, specification, and drawings. The sample of 150 is the production sample, and the sample of 200 is the periodic production sample. The lot shall be considered satisfactory if a sample of 150 meets the requirements of 3.2.3 and 3.3.3.1 through 3.3.3.4. The sample of 200 shall meet the requirements of 3.2.3, 3.3.3.1 through 3.3.3.4, 3.4.1.1, 3.4.1.2, 3.4.2.1, 3.4.6.1, and 3.4.6.2. The periodic production sample shall be allocated as shown in Figure 2. The periodic production sample is not required to meet 3.3.3.5.

4.3 Classification of Tests - The inspection and testing of the detonator assembly shall be classified as follows:

4.3.1 Production Tests - These tests are to be accomplished on each detonator or component being submitted for acceptance under the contract. Production tests shall be either performed by the manufacturer and witnessed by the Government Inspector, or these tests may be conducted by the Government Inspector. These tests are detailed in 4.5, except for paragraph 4.5.4.4.4.



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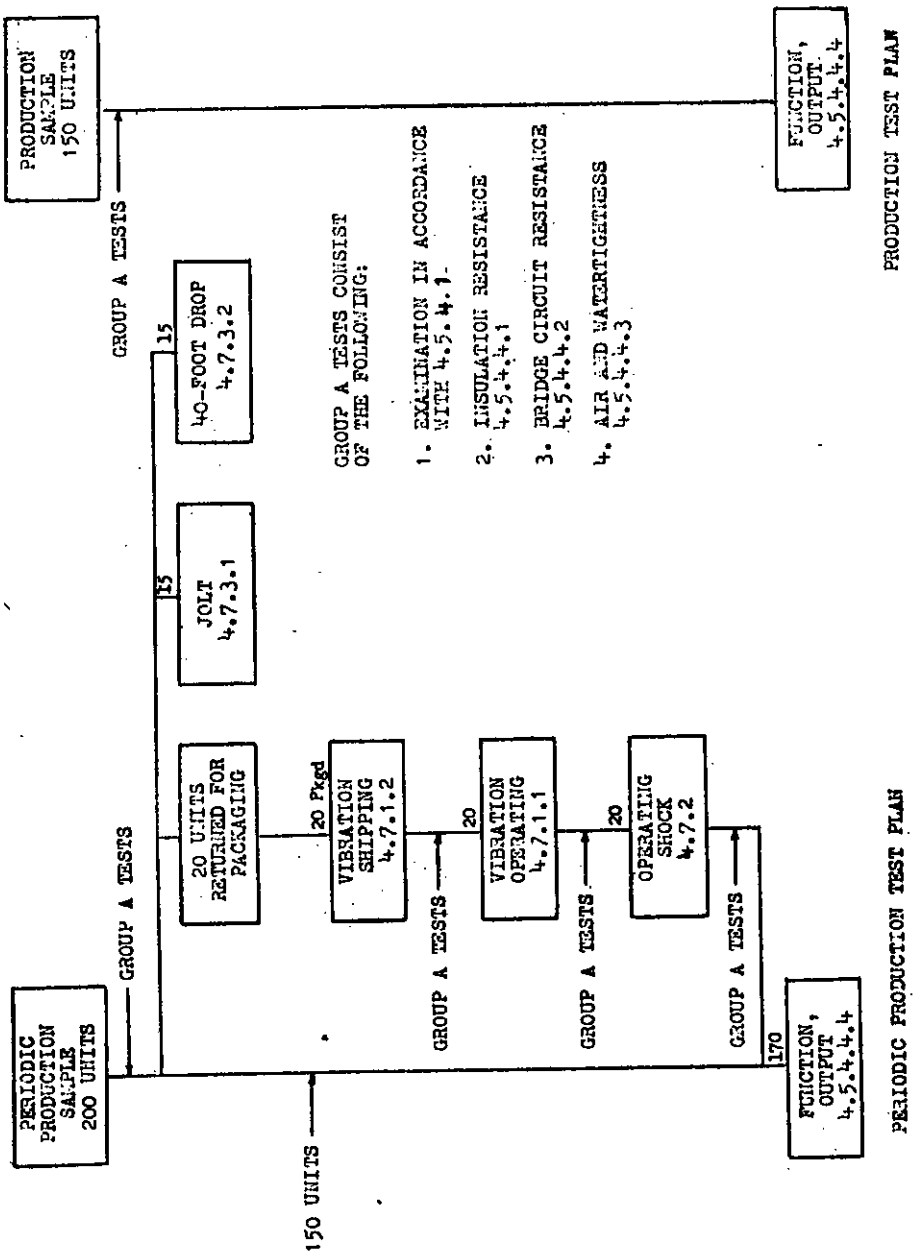



Figure 2



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4.3.2 Preproduction Tests - Preproduction tests are those which are accomplished on samples selected as specified in 4.2.1, which are representative of the production of detonator assemblies, to determine that production detonators will meet the requirements of this specification. These tests are detailed in 4.5 and 4.6. Failure of the sample to comply with these requirements will result in the rejection of the lot and cessation of production until determination of cause of failure is made and appropriate correction approved by the procuring activity.

4.3.3 Periodic Tests - Periodic tests are those which are accomplished on samples selected as specified in 4.2.4 to determine that the production meets the requirements of this specification. These tests are detailed in 4.5 and 4.7.

4.3.4 Tests After Delivery - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all of the tests specified herein as deemed necessary after delivery of the detonator assembly. Any defects discovered before, during, or after these tests, due to the contractor's failure to comply with drawings or specifications for this equipment, shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.4 Special Test Equipment - The following items of special test equipment are required to perform the acceptance tests set forth in this specification:

SA-2586902 Detonator Test Equipment

4.5 Production Tests

4.5.1 Responsibility - The contractor or Government Inspector shall conduct production tests as specified in 4.3.1 to assure that the detonator assembly or component is in compliance with the requirements of 3.2 and 3.3.



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4.5.2 Test Conditions - Unless otherwise specified, the detonator assemblies shall be subjected to production tests under the following conditions:

| | |
|---|---|
| Temperature: | Room ambient, +65 to +85 degrees Fahrenheit |
| Pressure: | 14.7 ± 3 psia |
| Vibration: | None |
| Humidity: | Relative humidity of 50 percent or less at +78 degrees Fahrenheit |

4.5.3 Test and Inspection Equipment and Facilities - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all production tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required production tests.

4.5.3.1 Calibration - Measuring equipment and instrumentation shall be calibrated as required by NAVORD INST 4355.1, 4355.28, and in accordance with the policies specified in the Standards Laboratory Information Manual issued by Bureau of Naval Weapons Representative, Pomona, California.

4.5.4 Test Procedures

4.5.4.1 Materials and Dimensions - Each detonator assembly shall be examined for conformance to the requirements of 3.2.



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4.5.4.2 Detonator Bridge Plug

4.5.4.2.1 Insulation Resistance Test (Bridge Plug) - The insulation resistance of each plug shall be measured between the shorted leads and the eyelet by means of Test Set, Mark 160 Mod 1, operated in accordance with instructions given in NAVORD OD 6852. The temperature and relative humidity during the test shall be less than +79 degrees Fahrenheit and 80 percent, respectively. To facilitate production, ambient conditions may exceed the prescribed limits for acceptance but rejection of material shall be based on tests conducted as prescribed. An acceptable assembly shall have a resistance of not less than 50 megohms when a potential of 500-volts DC is applied.

4.5.4.2.2 Circuit Resistance Test (Bridge Plug) - The bridge circuit resistance of each plug shall be measured with Test Set, Mark 238 Mod 0, operated similarly to the procedures described in NAVORD OD 6854. The testing current passing through the bridge circuit shall be less than 5 milliamperes. Bridged plugs having a resistance outside the limits shown on BuWep Dwg 2016961 shall not be further assembled, but shall not cause rejection of the lot.

4.5.4.3 Primer Holder and Bridge-Wire Assembly - The force required to seat the primer holder on the bridge-wire assembly shall be measured for each subassembly. When an assembly force of less than 10 pounds is required to seat the primer holder, the subassembly shall not be further assembled, but the failure shall not cause rejection of the lot.

4.5.4.4 Detonator Assembly

4.5.4.4.1 Insulation Resistance Test - A potential of 500-volts DC shall be applied between the contact prongs and the cup of each detonator. The apparatus for testing shall be Test Set, Mark 160 Mod 1, operated similarly to the procedures described in NAVORD OD 6852. The detonator shall be contained in a safety chamber during testing. The temperature and relative humidity during the test shall be less than +79 degrees Fahrenheit and 80 percent, respectively. To facilitate production, ambient conditions may exceed the prescribed limit for acceptance but rejection of material shall be based on tests conducted as prescribed. Detonators having an insulation resistance of less than 50 megohms shall be rejected.



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4.5.4.4.2 Bridge Circuit Resistance Test - The bridge circuit resistance of each detonator shall be measured using a current of less than 5 milliamperes. Measurements shall be made with Test Set, Mark 238 Mod 0. The test set operating procedures shall be similar to those described in NAVORD OD 6854. The detonator shall be enclosed in a suitable safety chamber during test. Detonators having a resistance outside the limits of BuWep Dwg 2016961 shall be rejected.

4.5.4.4.3 Air/Watertightness Test - Each detonator shall be immersed in water with a suitable wetting agent (Monsanto Sterol-AJ, or equal) and subjected to an absolute pressure of 2.5 inches of mercury for not less than 15 minutes followed by a pressure of 130 ± 5 psig for not less than 30 minutes. After the pressure is reduced to atmospheric, the detonator shall be removed from the water, air dried for 6 hours, and then weighed. An increase in weight in excess of 0.025 percent shall cause rejection of the detonator but shall not cause rejection of the lot. However, any detonator which has previously passed this test, which is subsequently discovered to be a leaker on retesting shall cause rejection of the lot.

4.5.4.4.4 Function/Output Test - One hundred fifty detonators from each lot of 650 or 170 detonators from each lot of 700, individually mounted on an aluminum wedge as specified in BuWep Dwg SA-2586902 shall be actuated (in a protective test chamber) by use of the electrical test circuit shown on BuWep Dwg SA-2586902. Any detonator which fails to function on the first discharge of the capacitor, or which fails to produce a crater to satisfy the requirements of 3.3.3.4 shall cause rejection of the lot from which selected.

4.6 Preproduction Tests

4.6.1 Vibration Tests

4.6.1.1 Vibration Test, Operating Condition - Forty-eight detonator assemblies rigidly mounted on a vibration table in the test block, BuWep Dwg SA-2586902, shall be subjected to vibration tests in accordance with Standard MIL-STD-167, Type I, in the range of 5 to 25 cycles-per-second, to meet the requirements of 3.4.1.1.



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4.6.1.2 Vibration Test, Shipping Condition - Forty-eight detonator assemblies, packaged and packed as specified in section 5 herein and rigidly mounted on a vibration table shall be vibrated at amplitudes equivalent to the maximum g's specified for a minimum of 2 minutes for each step of 2 cycles-per-second from the minimum capabilities of the equipment through the maximum frequency given in 3.4.1.2.

4.6.2 Shock Tests

4.6.2.1 Shock Test, Operating Condition - Forty-eight detonator assemblies, rigidly mounted in the test block shown on BuWep Dwg SA-2586902, shall be gradually subjected to a maximum acceleration of 3 g upward and 5 g downward (including gravity) and 4 g laterally. Any detonator which explodes or becomes unsafe to handle or use shall cause rejection of the lot.

4.6.2.2 Shock Test, Shipping Condition

4.6.2.2.1 Impact Test - Thirty-two detonators, packaged and packed as specified in section 5 herein, shall be rigidly mounted on the platform of a package tester. The platform shall be oscillated so that it moves in a circular path with a peak-to-peak amplitude of 1 inch. The frequency vibration (normally 4 to 5 cps) shall be set at the point where the pack jostles against the fence and momentarily leaves the platform by approximately 3/16 inch. The pack shall be vibrated on each of the six faces for 30 minutes.

4.6.2.2.2 Corner-Drop Test - Thirty-two detonators, packaged and packed as specified in section 5 herein, shall be dropped from a height of 30 inches onto a steel or concrete surface in such a manner that the corner of the container absorbs the full force of the fall. No ropes or other suspending media shall be attached to the package during free fall, and the 30 inches refers to the distance from the surface to the nearest corner of the container prior to the fall. All eight corners of the package shall be tested.



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4.6.3 Temperature and Humidity Test - Thirty detonators shall be subjected to the test specified in Standard MIL-STD-354.

4.6.4 Corrosive Atmosphere Test - Thirty detonators shall be subjected to the salt-spray test specified in Standard MIL-STD-356. One half of the detonators shall be mounted nose-up and one half nose-down.

4.6.5 Fungus Test - Twenty-eight detonators, without protective packaging and suspended in an upright position, shall be subjected to the Procedure I fungus test detailed in Specification MIL-E-5272.

4.6.6 Safety Tests

4.6.6.1 Jolt Test - Thirty detonator assemblies, mounted in the test block shown on BuWep Dwg SA-2586902, shall be subjected to the jolt test defined in Standard MIL-STD-350. When one or more detonators explode, become unsafe to handle, or require disposal of, as a result of the jolt test, the lot shall be rejected.

4.6.6.2 Forty-Foot Drop Test - Thirty detonators, mounted in the test block shown on BuWep Dwg SA-2586902, shall be subjected to the drop test indicated in Standard MIL-STD-352. When one or more detonators explode, become unsafe to handle, or require disposal of, as a result of the drop test, the lot shall be rejected.

4.6.6.3 Four-Foot Drop Test - Thirty unpackaged detonators shall be dropped from a height of 4 feet onto a steel or concrete surface. Any detonator which explodes or becomes unsafe to handle or use shall cause rejection of the lot. No suspending media shall be attached to the detonator during the free fall. The test shall be conducted behind suitable shielding, and the shorting clip, BuWep Dwg 2009330, shall be attached to each detonator.



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4.6.7 Function/Output/Time Test - Four hundred forty detonators from test plan of Figure 1, individually mounted on an aluminum wedge as specified in BuWep Dwg SA-2586902 shall be actuated (in a protective test chamber) by use of the electrical test circuit shown on BuWep Dwg SA-2586902. Any detonator which fails to function on the first discharge of the capacitor, within 600 microseconds after actuation, or which fails to produce a crater to satisfy the requirements of 3.3.3.4, shall cause rejection of the lot from which selected.

4.6.8 Sensitivity

4.6.8.1 Current - Thirty detonators shall be subjected to a steady electrical current of 100 milliamperes DC through the bridge wire for 24 hours and shall not fire.

4.6.8.2 Radio Frequency Test - Thirty detonator assemblies, suitably installed under simulated operating conditions, shall be subjected to 10-seconds exposure, at a 10-foot distance, to radar having a nominal power of 3 megawatts over a frequency range covered in the "L", "S", "C", and "X" bands, and shall not fire.

4.7 Periodic Production Tests

4.7.1 Vibration Tests

4.7.1.1 Operating Condition - Twenty detonators shall be tested as specified in 4.6.1.1.

4.7.1.2 Shipping Condition - Twenty packaged detonators shall be tested as specified in 4.6.1.2.

4.7.2 Shock, Operating Condition - Twenty detonators shall be tested as specified in 4.6.2.1.

4.7.3 Safety Tests

4.7.3.1 Jolt Test - Fifteen detonators shall be tested as specified in 4.6.6.1.

4.7.3.2 Forty-Foot Drop Test - Fifteen detonators shall be tested as specified in 4.6.6.2.



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4.8 Rejection - Except as noted in 4.5.4.2.1, 4.5.4.2.2, 4.5.4.3, 4.5.4.4.3, and 4.8.1, failure of the individual detonator assembly to meet any of the requirements and tests of this specification shall be considered cause for rejection of the lot.

4.8.1 Failure During Periodic Tests - Failure of the sample to pass the tests of 4.7 shall not cause rejection of the lot. However, production shall cease until determination of the cause of failure is made, appropriate corrections are approved by the contracting activity, and preproduction tests of 4.6 are passed.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Preservation and packaging shall be Level A or Level C, as follows:

5.1.1 Level A - Level A packaging shall be such as to assure adequate protection to prevent corrosion, deterioration, or physical damage during shipment, handling, indeterminate storage, and world-wide distribution.

5.1.1.1 Packaging - The detonator assemblies shall be clean and dry and fitted with individual shorting clips. The detonators shall be inserted in a square, flat, wood holder, in snug (not tight) fitting holes on 1-inch centers, prongs up, in a group of four detonators to a holder. Each holder with detonators shall be closely wrapped in aluminum foil conforming to Specification MIL-A-148. The wrap shall be placed in a tight-fitting fiberboard carton, full telescope style, conforming to Specification PPP-B-676. Closure shall be in accordance with good commercial practice.

5.1.1.2 Intermediate Packaging

5.1.1.2.1 Sixteen Packages per Carton - The unit package specified in 5.1.1.1 shall be placed in a tight-fitting, intermediate package in groups of 16 packages to a carton (64 detonators). The carton for the intermediate packaging shall be fiberboard, full telescope style, conforming to Specification PPP-B-636. Closure shall be in accordance with good commercial practice.



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5.1.1.2.2 Two Packages per Carton - The unit package specified in 5.1.1.1 shall be placed in a tight-fitting, intermediate package in groups of 2 packages to a carton (8 detonators). The carton for the intermediate packaging shall be fiberboard, full telescope style, conforming to Specification PPP-B-636. Closure shall be in accordance with good commercial practice.

5.1.2 Level C - Cleaning, drying, preservation, and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing - Packing of the detonators shall be Level A or Level C, as follows:

5.2.1 Level A - The detonators, packaged as specified in 5.1.1.1 and 5.1.1.2.1 or 5.1.1.2.2, shall be packed in a quantity not to exceed 384 detonators to a shipping container. The shipping container shall be a wood box conforming to Specification PPP-B-621. The method of packing for shipment shall conform to Section 73.66 (g) (1) of the Interstate Commerce Commission Regulations.

5.2.2 Level C - The detonators packaged as specified in 5.1.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with the Uniform Freight Classification Rules or other common-carrier regulations applicable to the mode of transportation.

5.3 Marking - Unit packages, intermediate packages and shipping containers shall be marked in accordance with the Code of Regulation 49 CFR 71-78 and Standard MIL-STD-129.



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6. NOTES

6.1 Ordering Data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Exceptions to this specification or any specification, document, or drawing referenced herein.

(c) Selection of applicable levels of preservation and packaging and quantity required per intermediate package.

6.2 General Safety Precautions - The loading, assembly, and handling of explosives, ammunition, ammunition components, fuzes, detonators, primers, tracers, boosters, pyrotechnics, etc., and subassemblies thereof, involve hazardous operations and therefore require suitable explosives safety precautions. Use of this specification shall not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. Listed below are the minimum provisions which a contractor or manufacturer (loader) of the above items should observe in order to fulfill his responsibility for safety. At Bureau of Naval Weapons, Navy Department, and other Government plants, these provisions are mandatory. Such other warnings and precautions pertinent to the operational effectiveness or safety during use of loading of the specified item, are included in the detail provisions of the specification.

6.2.1 All loading operations should be conducted in a neat and orderly manner.

6.2.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations, such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.



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6.2.3 Personnel handling detonators and related parts which effect functioning, should in so far as practicable avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.2.4 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.2.5 All explosives and completely or partially loaded items should be stored in suitable magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with intra-plant distances when outside of loading rooms. For Navy explosives loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives, shall apply.

6.2.6 Proper care should be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In this connection, the following precautionary measures should be observed.

6.2.6.1 Employ properly proportioned and properly located protective barricades, screens, or shields at all required points.

6.2.6.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.2.6.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground. electrostatic charges.



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6.2.6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing should be of a type to minimize the accumulation of static charges. Fabrics, such as silk and nylon, which promote static generation should be avoided. Additional grounding devices, such as grounded bracelets for workers, should be employed where operations are conducted with items which are usually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low-energy type electric primers, detonators and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare noninsulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency heating apparatus, etc. Where necessary for safety, humidity (see 6.2.4) of work rooms should be increased as required to lessen electrostatic effects without excessive moisture absorption.

6.2.6.5 Protect all explosive operations from effects of electric current originating from equipment, such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding, separation, or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electronic soldering iron through a grounded contact. All electric type primers, detonators, or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips, or other devices should be employed during manufacturing operations involving electric primers, detonators or squibs.



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6.2.6.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning, and associated industrial hazards.

6.2.6.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling, and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.2.6.8 Enforce good housekeeping and maintain effective policing, inspection, and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, equipment, and the items loaded. Clean up any spilled material immediately.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

| ORGANIZATION *(Of submitter)* | CITY AND STATE |
|---|---|

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

(NAVSHIPS OVPRT 12-66)

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